

Problem

Obtain the inverse Laplace transforms of the following functions:

(a) $X(s) = \frac{3}{s^2(s+2)(s+3)}$

(b) $Y(s) = \frac{2}{s(s+1)^2}$

(c) $Z(s) = \frac{5}{s(s+1)(s^2+6s+10)}$

Step-by-step solution

Step 1 of 8

a)

The function $X(s)$ is,

$$X(s) = \frac{3}{s^2(s+2)(s+3)}$$

Use partial fractions to simplify the function $Z(s)$.

$$\frac{3}{s^2(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{s+3} \dots\dots (1)$$

$$3 = As(s+2)(s+3) + B(s+2)(s+3) + Cs^2(s+3) + Ds^2(s+2) \dots\dots (2)$$

$$3 = s^3(A+C+D) + s^2(5A+B+3C+2D) + s(6A+5B) + 6B \dots\dots (3)$$

Compare the coefficient of constant terms in the equation (3).

$$6B = 3$$

$$B = \frac{1}{2}$$

Step 2 of 8

Substitute $s = -2$ in the equation (2).

$$3 = As(-2+2)(-2+3) + B(-2+2)(-2+3) + C(-2)^2(-2+3) + D(-2)^2(-2+2)$$

$$3 = 0 + 0 + C(-2)^2(-2+3) + 0$$

$$C = \frac{3}{4}$$

Substitute $s = -3$ in the equation (2).

$$3 = A(-3)(-3+2)(-3+3) + B(-3+2)(-3+3) + C(-3)^2(-3+3) + D(-3)^2(-3+2)$$

$$3 = 0 + 0 + 0 + D(9)(-1)$$

Compare

$$D = -\frac{1}{3}$$

the coefficient of s^3 in the equation (3) and substitute $C = \frac{3}{4}$ and $D = -\frac{1}{3}$.

$$A + C + D = 0$$

$$A + \frac{3}{4} - \frac{1}{3} = 0$$

$$A = -\frac{5}{12}$$

Step 3 of 8

Substitute $A = -\frac{5}{12}$, $B = \frac{1}{2}$, $C = \frac{3}{4}$ and $D = -\frac{1}{3}$ in equation (1).

$$\frac{3}{s^2(s+2)(s+3)} = -\frac{5}{12s} + \frac{1}{2s^2} + \frac{3}{4(s+2)} - \frac{1}{3(s+3)}$$

Take inverse Laplace transform to the above equation.

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{3}{s^2(s+2)(s+3)}\right] &= \mathcal{L}^{-1}\left(-\frac{5}{12s}\right) + \mathcal{L}^{-1}\left(\frac{1}{2s^2}\right) + \mathcal{L}^{-1}\left(\frac{3}{4(s+2)}\right) \\ &\quad + \mathcal{L}^{-1}\left[-\frac{1}{3(s+3)}\right] \\ &= -\frac{5}{12}u(t) + \frac{t}{2}u(t) + \frac{3}{4}e^{-2t}u(t) - \frac{1}{3}e^{-3t}u(t) \\ &= \left(-\frac{5}{12} + \frac{t}{2} + \frac{3}{4}e^{-2t} - \frac{1}{3}e^{-3t}\right)u(t)\end{aligned}$$

Therefore, the Laplace transform of the function $X(s)$ is $\boxed{\left(-\frac{5}{12} + \frac{t}{2} + \frac{3}{4}e^{-2t} - \frac{1}{3}e^{-3t}\right)u(t)}$.

Step 4 of 8

b)

The function $Y(s)$ is,

$$Y(s) = \frac{2}{s(s+1)^2}$$

Use partial fractions to simplify the function $Z(s)$.

$$\frac{2}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} \dots\dots (4)$$

$$2 = A(s+1)^2 + Bs(s+1) + Cs \dots\dots (5)$$

$$2 = s^2(A+B) + s(2A+B+C) + A \dots\dots (6)$$

Compare the coefficient of constant terms in the equation (6).

$$A = 2$$

Substitute $s = -1$ in the equation (5).

$$2 = A(-1+1)^2 + B(-1)(-1+1) + C(-1)$$

$$2 = 0 + 0 + C(-1)$$

$$C = -2$$

Compare coefficient of s^2 terms in equation (6) and replace $A = 2$.

$$A + B = 0$$

$$B = -2$$

Substitute $A = 2$, $B = -2$ and $C = -2$ in equation (4).

$$\frac{2}{s(s+1)^2} = \frac{2}{s} - \frac{2}{s+1} - \frac{2}{(s+1)^2}$$

Take inverse Laplace transform to the above equation.

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{2}{s(s+1)^2}\right] &= \mathcal{L}^{-1}\left(\frac{2}{s}\right) - \mathcal{L}^{-1}\left(\frac{2}{s+1}\right) - \mathcal{L}^{-1}\left(\frac{2}{(s+1)^2}\right) \\ &= 2u(t) - 2e^{-t}u(t) - 2te^{-t}u(t) \\ &= 2(1 - e^{-t} - te^{-t})u(t) \end{aligned}$$

Therefore, the Laplace transform of the function $Y(s)$ is $\boxed{2(1 - e^{-t} - te^{-t})u(t)}$.

Step 5 of 8

c)

The function $Z(s)$ is,

$$Z(s) = \frac{5}{s(s+1)(s^2+6s+10)}$$

Use partial fractions to simplify the function $Z(s)$.

$$\frac{5}{s(s+1)(s^2+6s+10)} = \frac{A}{s} + \frac{B}{s+1} + \frac{Cs+D}{s^2+6s+10} \dots\dots (7)$$

$$5 = A(s+1)(s^2+6s+10) + Bs(s^2+6s+10) + (Cs+D)(s^2+s) \dots\dots (8)$$

$$5 = s^3(A+B+C) + s^2(7A+6B+C+D) + s(16A+10B+D) + 10A \dots\dots (9)$$

Step 6 of 8

Compare coefficient of constant terms in equation (9).

$$10A = 5$$

$$A = \frac{1}{2}$$

Substitute $s = -1$ in the equation (8).

$$5 = \left[A(-1+1)(-1^2 + 6(-1) + 10) + B(-1)((-1)^2 + 6(-1) + 10) \right] \\ + (C(-1) + D)((-1)^2 - 1)$$

$$5 = 0 + B(-1)(1 - 6 + 10) + 0$$

$$B = -1$$

Step 7 of 8

Compare coefficient of s^3 terms in equation (9) and replace $A = \frac{1}{2}$ and $B = -1$.

$$A + B + C = 0$$

$$\frac{1}{2} - 1 + C = 0$$

$$C = \frac{1}{2}$$

Compare coefficient of s^2 terms in equation (9) and replace $A = \frac{1}{2}$, $B = -1$ and $C = \frac{1}{2}$.

$$7A + 6B + C + D = 0$$

$$7\left(\frac{1}{2}\right) + 6(-1) + \frac{1}{2} + D = 0$$

$$D = 2$$

Substitute $A = \frac{1}{2}$, $B = -1$, $C = \frac{1}{2}$ and $D = 2$ in equation (7).

$$\frac{5}{s(s+1)(s^2+6s+10)} = \frac{1}{2s} - \frac{1}{s+1} + \left(\frac{\frac{1}{2}s+2}{s^2+6s+10} \right) \\ = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2} \left(\frac{s+4}{s^2+6s+10} \right)$$

Step 8 of 8

Take inverse Laplace transform to the above equation.

$$\begin{aligned}
 \mathcal{L}^{-1}\left[\frac{5}{s(s+1)(s^2+6s+10)}\right] &= \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{1}{2}\left(\frac{s+4}{s^2+6s+10}\right)\right) \\
 &= \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) + \frac{1}{2}\mathcal{L}^{-1}\left(\left(\frac{s+3+1}{(s+3)^2+1}\right)\right) \\
 &= \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) \\
 &\quad + \frac{1}{2}\mathcal{L}^{-1}\left(\frac{s+3}{(s+3)^2+1}\right) + \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{(s+3)^2+1}\right) \\
 &= \frac{1}{2}u(t) - e^{-t}u(t) + \frac{1}{2}\cos te^{-3t}u(t) + \frac{1}{2}\sin te^{-3t}u(t) \\
 &= \frac{1}{2}(1 - 2e^{-t} + \cos te^{-3t} + \sin te^{-3t})u(t)
 \end{aligned}$$

Therefore, the Laplace transform of the function $Z(s)$ is

$$\boxed{\frac{1}{2}(1 - 2e^{-t} + \cos te^{-3t} + \sin te^{-3t})u(t)}.$$